

Module 6 :: Latches and Flip-Flops

Practice Problems

1. Draw the state (truth) tables of: (i) S-R flip-flop, (ii) D flip-flop, (iii) T flip-flop, (iv) J-K flip-flop.
2. Show how you can carry out the following conversions:
 - a) S-R flip-flop to D flip-flop
 - b) S-R flip-flop to T flip-flop
 - c) S-R flip-flop to J-K flip-flop
 - d) J-K flip-flop to D flip-flop
 - e) J-K flip-flop to T flip-flop
 - f) D flip-flop to S-R flip-flop
 - g) T flip-flop to J-K flip-flop
3. Draw the schematic diagram using NAND gates for:
 - a) S-R master-slave flip-flop
 - b) J-K master-slave flip-flop
4. What is a race condition in an S-R flip-flop? Explain how this can be avoided in a J-K flip-flop.
5. Show how a T flip-flop can divide the frequency of an input rectangular pulse train by 2, and generate a waveform with exactly 50% duty cycle.